SMART TEMPERATURE AND GAS DETECTION SYSTEM USING ARDUINO UNO

A Project Report

Submitted in partial fulfilment of the requirements for the award of the degree of

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ELECTRICAL AND ELECTRONICS ENGINEERING

Submitted by

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This project work has been carried out under my supervision and has not been submitted, either in part or in full, for the award of any other degree or diploma of this or any other institution.

Signature of Faculty Guide   Signature of Head of the Department

Submitted for the Viva-Voce examination held on ____________.

Internal Examiner   External Examiner

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KARTHIKEYAN M

ABSTRACT

Gas leakage and abnormal rise in temperature are two of the most common causes of fire accidents and equipment damage in homes, laboratories, and small-scale industries. Early detection of such hazardous conditions can significantly reduce the risk of accidents and protect both human life and property. This project presents the design and implementation of a Smart Temperature and Gas Detection System built around the Arduino UNO microcontroller.

The system uses a DHT11 Temperature and Humidity Sensor to continuously monitor the surrounding temperature and humidity, and an MQ Gas Sensor to detect the presence of combustible gases such as LPG, methane, and smoke. The Arduino UNO reads both sensor values in real time and compares them against predefined threshold values of 28°C for temperature and 400 for the gas sensor reading.

Based on the comparison, the system displays one of four distinct status messages on a 16×2 I2C LCD display — SAFE ZONE, TEMP WARNING, GAS WARNING, or DANGER — depending on whether the temperature, the gas concentration, both, or neither exceeds the set threshold. Whenever an unsafe condition is detected, an LED and an active buzzer are triggered automatically to provide instant visual and audible alerts.

Unlike many conventional monitoring systems that provide only a single generic alarm, the proposed system clearly differentiates between the type of hazard detected, allowing the user to respond appropriately and quickly. The complete hardware was designed and simulated using Cirkit Designer, and the program was developed using Embedded C in the Arduino IDE. The prototype was assembled on a breadboard and tested under different environmental conditions, and all four operating states were verified successfully.

The results confirm that the developed system is simple, low-cost, reliable, and easy to implement, making it well suited for educational demonstrations as well as basic safety monitoring applications in homes, kitchens, and laboratories. The project also lays a foundation for future enhancement through IoT connectivity, mobile alerts, and automatic hazard-control mechanisms.

Keywords: Arduino UNO, DHT11 Sensor, MQ Gas Sensor, LCD Display, Gas Leakage Detection, Temperature Monitoring, Embedded System, Safety Alert System.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Safety monitoring has become an important requirement in homes, industries, laboratories, and commercial buildings. Gas leakage and abnormal rise in temperature are among the major causes of accidents, fire hazards, and equipment damage. Detecting these conditions at an early stage helps reduce risks and protect both people and property.

This project presents a Smart Temperature and Gas Detection System developed using the Arduino UNO microcontroller. The system continuously monitors environmental conditions using a DHT11 Temperature and Humidity Sensor and an MQ Gas Sensor. The measured values are processed by the Arduino in real time.

Whenever the measured temperature exceeds 28°C or the gas sensor value exceeds 400, the system immediately activates a buzzer and an LED to alert the user. At the same time, a 16×2 I2C LCD displays an appropriate warning message indicating the detected condition.

Unlike conventional monitoring systems that only indicate a general alarm, this project identifies different operating conditions by displaying separate messages such as SAFE ZONE, TEMP WARNING, GAS WARNING, and DANGER. This enables users to quickly understand the source of the detected hazard and take suitable action.

The system is simple, low-cost, reliable, and suitable for educational purposes as well as basic environmental safety applications.

1.2 Problem Statement

Gas leakage and abnormal temperature can cause serious accidents if they are not detected at an early stage. Many low-cost monitoring systems only provide a common alarm without identifying the actual source of the problem. This makes it difficult for users to understand whether the warning is caused by high temperature, gas leakage, or both conditions occurring together.

To overcome this limitation, this project develops a monitoring system capable of identifying individual warning conditions while providing real-time environmental monitoring and immediate alerts, so that the user can respond according to the specific hazard detected.

1.3 Objectives

The main objectives of this project are as follows:

- To monitor surrounding temperature and humidity using the DHT11 sensor.
- To detect harmful gas leakage using the MQ Gas Sensor.
- To continuously display sensor values on a 16×2 LCD display.
- To activate an LED and buzzer whenever unsafe conditions are detected.
- To display different warning messages based on the detected condition.
- To develop a reliable and low-cost embedded safety monitoring system using Arduino UNO.

1.4 Scope of the Project

The developed system is intended for basic environmental monitoring and safety applications. It continuously measures temperature and gas concentration and alerts users whenever the measured values exceed predefined threshold levels.

The project can be applied in the following areas:

- Residential homes
- Kitchens
- Laboratories
- Small-scale industries
- Chemical storage areas
- Educational laboratories

Future improvements such as IoT connectivity, mobile notifications, cloud monitoring, and automatic ventilation control can further increase the capabilities of the system.

1.5 Applications

The major features and application areas of the developed system are summarised below:

- Real-time temperature and humidity monitoring
- Real-time gas leakage detection
- Continuous LCD display of sensor readings
- Audible warning through buzzer
- Visual indication using LED
- Individual warning messages for different hazard conditions
- Low-cost implementation using easily available components
- Easy hardware integration and maintenance
- Arduino-based embedded design suitable for student projects
- Reliable operation for educational and basic safety applications

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Environmental monitoring systems have become increasingly important in recent years due to the growing need for safety in residential, commercial, and industrial environments. Gas leakage and abnormal temperature are among the leading causes of fire accidents and equipment failures. Various researchers have proposed monitoring systems using microcontrollers and sensors to improve safety and provide early warning.

Recent developments in embedded systems have made it possible to design compact, low-cost, and reliable monitoring devices capable of continuously observing environmental conditions. Arduino-based systems are widely adopted because of their simplicity, flexibility, and large open-source ecosystem.

This chapter reviews several existing monitoring systems and highlights their limitations, which motivated the development of the proposed Smart Temperature and Gas Detection System.

2.2 Existing System

Several temperature and gas monitoring systems have been developed using different sensing technologies and communication methods.

Some systems use only temperature sensors to detect overheating, while others rely solely on gas sensors for leakage detection. These systems provide only partial monitoring and cannot identify multiple hazards simultaneously.

Recent IoT-based monitoring systems transmit sensor data to cloud platforms for remote observation. Although these systems offer additional features, they increase the overall cost, require internet connectivity, and involve more complex hardware and software configurations.

Many low-cost systems also provide only a general alarm without specifying whether the alert was caused by temperature, gas leakage, or both.

2.3 Limitations of Existing Systems

The following limitations were identified from the review of previous systems:

- Most systems monitor only one environmental parameter.
- Many projects provide only a general alarm indication.
- Some systems require continuous internet connectivity.
- IoT-based solutions increase implementation cost.
- Lack of clear LCD messages makes fault identification difficult.
- Limited user feedback during abnormal conditions.
- Poor differentiation between temperature-related and gas-related hazards.

These limitations reduce the effectiveness of existing safety monitoring systems in practical applications.

2.4 Proposed System

The proposed Smart Temperature and Gas Detection System overcomes several limitations of existing systems by integrating both temperature and gas sensing into a single embedded platform.

The system continuously monitors environmental conditions using a DHT11 Temperature and Humidity Sensor together with an MQ Gas Sensor. The Arduino UNO processes sensor readings and compares them with predefined threshold values.

Unlike conventional systems, the proposed design provides four different operating states, namely SAFE ZONE, TEMP WARNING, GAS WARNING, and DANGER. This approach enables users to immediately identify the source of the warning and respond accordingly.

The LCD display continuously provides real-time sensor information, while the LED and buzzer provide instant visual and audible alerts whenever unsafe conditions are detected. The system is designed to be simple, economical, reliable, and suitable for educational as well as practical safety monitoring applications.

2.5 Comparison Table

Table 2.1 compares the features of existing monitoring systems with the proposed system.

Feature	Existing Systems	Proposed System
Temperature Monitoring	Yes	Yes
Gas Detection	Some Systems	Yes
Humidity Display	Limited	Yes
Separate Warning Messages	No	Yes
LCD Display	Limited	Yes
LED Alert	Yes	Yes
Buzzer Alert	Yes	Yes
Arduino Based	Some Systems	Yes
Low Cost	Moderate	Yes
Easy Implementation	Moderate	Yes

Table 2.1 Comparison of Existing and Proposed System

From the above comparison, it is evident that the proposed system offers a more complete and user-friendly monitoring solution compared to the majority of existing low-cost systems, while still remaining economical and easy to implement.

CHAPTER 3

HARDWARE DESCRIPTION

This chapter describes the hardware components used in the development of the Smart Temperature and Gas Detection System. Each component is explained along with its function in the overall system and its important technical specifications.

3.1 Arduino UNO

The Arduino UNO is the main controller used in this project. It is based on the ATmega328P microcontroller and provides sufficient digital and analog input/output pins to interface with sensors and output devices. It reads the sensor values, compares them with the predefined threshold values, and controls the LCD, LED, and buzzer accordingly.

Parameter	Specification
Microcontroller	ATmega328P
Operating Voltage	5 V
Digital I/O Pins	14
Analog Input Pins	6
Flash Memory	32 KB
SRAM	2 KB
Clock Speed	16 MHz

3.2 DHT11 Temperature and Humidity Sensor

The DHT11 sensor is used to measure the ambient temperature and humidity of the surroundings. It contains a capacitive humidity-sensing element and an NTC temperature sensor, and it sends calibrated digital data directly to the Arduino, which eliminates the need for additional signal conditioning circuitry.

Parameter	Specification
Operating Voltage	3.3 V – 5 V
Temperature Range	0°C to 50°C
Humidity Range	20% – 90%
Accuracy	±2°C

Table 3.1 DHT11 Sensor Specifications

3.3 MQ Gas Sensor

The MQ Gas Sensor is used to detect combustible gases and smoke present in the surrounding environment. It produces an analog output voltage that is proportional to the concentration of gas detected, and this analog value is continuously monitored by the Arduino through its analog input pin.

Parameter	Specification
Operating Voltage	5 V
Output Type	Analog
Sensitivity	High
Response Time	Fast
Detectable Gases	LPG, Methane, Smoke

Table 3.2 MQ Gas Sensor Specifications

3.4 LCD Display

A 16×2 I2C LCD display is used in this project to present system information to the user. It continuously displays the current temperature, the gas sensor value, and the system status such as SAFE ZONE, TEMP WARNING, GAS WARNING, or DANGER. The I2C interface reduces wiring complexity considerably by using only two communication lines, namely SDA and SCL, instead of the usual eight to ten data lines required by a standard parallel LCD.

3.5 LED

The LED provides a quick visual indication of the current operating condition of the system. The LED remains OFF during normal operation and turns ON automatically whenever the temperature or gas reading crosses its respective threshold value, thereby giving the user an immediate visual cue of the abnormal condition.

3.6 Buzzer

An active buzzer is used to provide an audible warning during unsafe conditions. Whenever the temperature or gas value exceeds the set threshold, the buzzer is switched on automatically along with the LED, allowing the user to recognise a hazardous situation even without directly observing the LCD display.

3.7 Power Supply

The complete circuit is powered through the USB cable connected to the Arduino UNO, which supplies a regulated 5 V required by the microcontroller, the LCD module, and the sensors. In the absence of a USB connection, the board can alternatively be powered using a 9 V DC adapter connected to its onboard barrel jack, which is internally regulated to the required 5 V using the Arduino's own voltage regulator circuitry.

A stable power supply is essential for accurate sensor readings, since fluctuations in supply voltage can directly affect the analog output of the MQ Gas Sensor and the timing accuracy of the DHT11 sensor. All ground connections of the sensors, LCD, LED, and buzzer are connected to a common ground rail on the breadboard to avoid floating references.

3.8 Pin Configuration

Component	Arduino Pin
DHT11 Data Pin	D4
MQ Gas Sensor Output	A0
LED	D8
Buzzer	D9
LCD SDA	A4
LCD SCL	A5

Table 3.3 Pin Configuration

CHAPTER 4

SOFTWARE DESCRIPTION

4.1 Arduino IDE

The Arduino Integrated Development Environment (IDE) was used to write, compile, and upload the program to the Arduino UNO board. The IDE provides a simple text editor along with a compiler and uploader, and it supports a large number of open-source libraries that simplify communication with external hardware such as sensors and displays.

The Arduino IDE was chosen for this project because of its beginner-friendly interface, wide community support, and compatibility with the ATmega328P microcontroller used on the Arduino UNO.

4.2 Embedded C

The program for this project was written in Embedded C, which is the standard programming language supported by the Arduino IDE. Embedded C allows direct control over the microcontroller's pins, timers, and peripherals, and it is well suited for real-time applications such as continuous sensor monitoring.

The program is structured into two main functions, setup() and loop(). The setup() function initialises the LCD, the DHT11 sensor, the serial communication, and the input/output pin modes. The loop() function executes repeatedly and performs the actual sensing, comparison, and output control operations.

4.3 Circuit Explanation

The Arduino UNO acts as the central controller of the system. It receives sensor data from the DHT11 Temperature and Humidity Sensor and the MQ Gas Sensor, processes the readings, compares them with predefined threshold values, and controls the output devices accordingly. The overall architecture of the system is illustrated in Figure 4.1.

The DHT11 sensor's data pin is connected to digital pin D4 of the Arduino, while the MQ Gas Sensor's analog output is connected to analog pin A0. The 16×2 I2C LCD is connected through the SDA and SCL lines to analog pins A4 and A5 respectively. The LED and buzzer are connected to digital pins D8 and D9, which are configured as outputs.

Figure 4.1 Block Diagram of the Proposed System

Power Supply → Arduino UNO → [DHT11 Sensor, MQ Gas Sensor, 16×2 LCD Display] → [LED, Buzzer]

The Arduino UNO forms the central node of the block diagram. It receives inputs continuously from the DHT11 sensor and the MQ gas sensor, and simultaneously drives the LCD display and the two output alert devices, the LED and the buzzer, based on the internally computed system status.

4.4 Working Principle

The system operates on a straightforward comparison-based logic. The DHT11 sensor continuously measures the ambient temperature and humidity, while the MQ Gas Sensor continuously measures the concentration of combustible gases in the surrounding air. Both readings are transmitted to the Arduino UNO, where a logical comparison is performed against the predefined threshold values of 28°C for temperature and 400 for the gas sensor reading.

If both readings remain within safe limits, the system continues normal operation and displays SAFE ZONE on the LCD. If either or both values exceed their respective thresholds, the Arduino immediately activates the LED and buzzer and updates the LCD with the appropriate warning message, namely TEMP WARNING, GAS WARNING, or DANGER.

4.5 Flowchart

Figure 4.2 presents the flowchart of the program using standard flowcharting symbols. The oval blocks represent the start of the process, the parallelogram blocks represent input and output operations, the rectangular blocks represent processing steps, and the diamond blocks represent decision points at which the temperature and gas readings are compared with their threshold values.

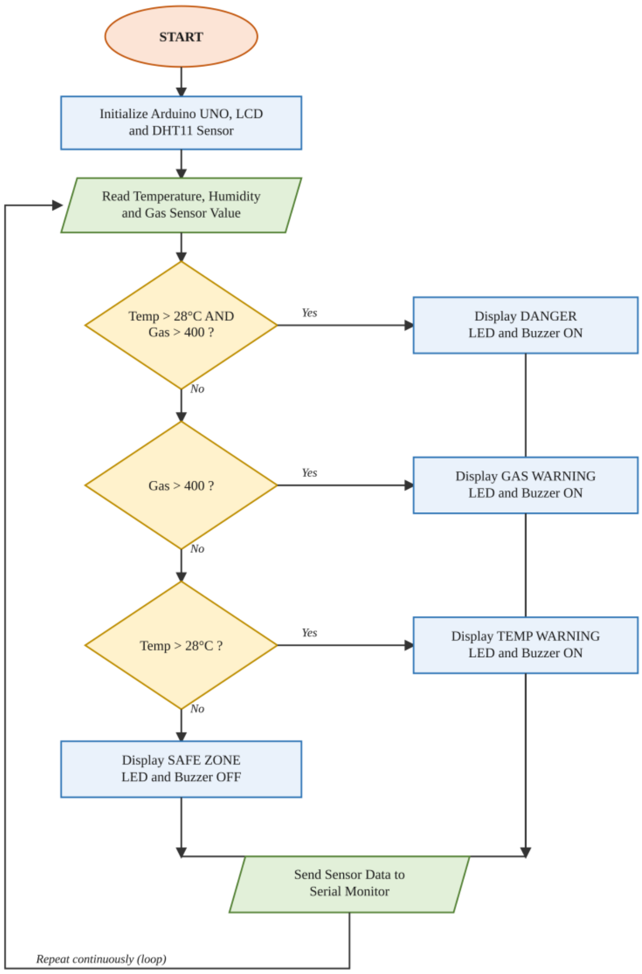

Figure 4.2 Flowchart of System Operation

As shown in the flowchart, the program first initialises the hardware and then continuously reads the sensor values. The readings are checked against the threshold conditions in order of priority — the combined danger condition is checked first, followed by the gas warning condition, and then the temperature warning condition — so that the most critical hazard is always reported first. If none of the conditions are satisfied, the system reports SAFE ZONE. After every decision, the sensor values are written to the Serial Monitor and the loop repeats continuously, giving the system its real-time monitoring behaviour.

4.6 Algorithm

The complete operation of the system follows the algorithm given below:

Step 1: Initialise Arduino UNO and LCD.

Step 2: Initialise the DHT11 sensor.

Step 3: Read temperature and humidity.

Step 4: Read the gas sensor value.

Step 5: Compare the values with the threshold limits.

Step 6: If Gas > 400, display GAS WARNING and turn ON the LED and buzzer.

Step 7: If Temperature > 28°C, display TEMP WARNING and turn ON the LED and buzzer.

Step 8: If both conditions are true simultaneously, display DANGER and turn ON the LED and buzzer.

Step 9: Otherwise, display SAFE ZONE and continue monitoring.

4.7 Program Description

The software was developed using the Arduino programming language within the Arduino IDE. The program performs the following tasks continuously inside the loop() function, ensuring real-time monitoring:

- Reads temperature and humidity from the DHT11 sensor.
- Reads gas concentration from the MQ Gas Sensor.
- Compares sensor readings with predefined threshold values.
- Displays current sensor values on the LCD.
- Generates warning messages during unsafe conditions.
- Activates the LED and buzzer whenever necessary.
- Sends the sensor readings to the Serial Monitor for verification.

Two external libraries were used to simplify the development process: the LiquidCrystal_I2C library, which controls the 16×2 LCD display over the I2C bus, and the DHT library, which handles the timing-sensitive communication protocol required to read data from the DHT11 sensor.

The threshold values used for comparison are summarised in Table 4.1 below.

Parameter	Threshold Value
Temperature	28°C
Gas Sensor Reading	400

Table 4.1 Threshold Configuration

The Arduino compares every new sensor reading against these values during each iteration of the loop. If either threshold is exceeded, the corresponding warning system is activated immediately, ensuring that the response of the system remains practically instantaneous from the user's point of view.

CHAPTER 5

HARDWARE IMPLEMENTATION

5.1 Hardware Setup

The hardware implementation consists of connecting all sensors and output devices to the Arduino UNO on a breadboard. The DHT11 sensor is connected to digital pin D4 for measuring temperature and humidity. The MQ Gas Sensor is connected to analog pin A0 to measure gas concentration. The 16×2 I2C LCD is connected through the SDA and SCL communication lines to display system information, and the LED and buzzer are connected to digital pins D8 and D9 respectively for visual and audible alerts.

The complete hardware prototype, assembled on a breadboard along with the Arduino UNO, sensors, and the LCD module, is shown in Figure 5.1.

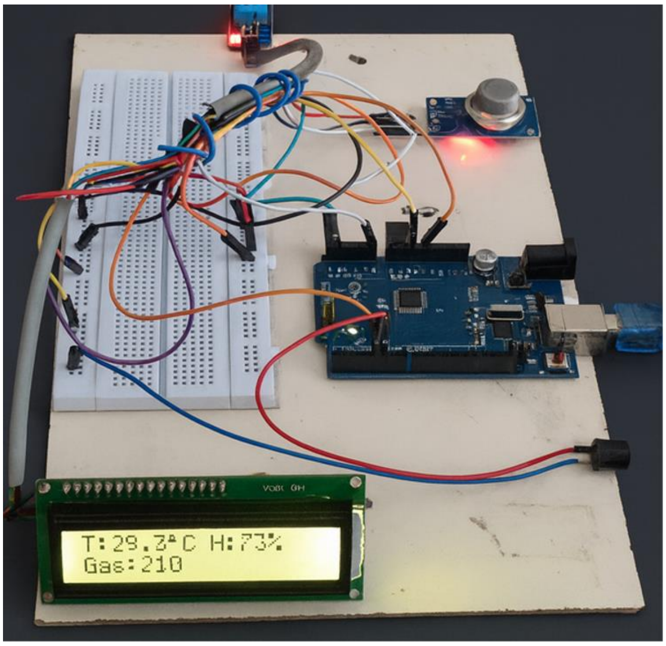

Figure 5.1 Hardware Prototype Setup

5.2 Circuit Diagram Explanation

The complete circuit was first designed and verified using Cirkit Designer before assembling the physical hardware. The simulation confirmed correct pin connections, proper sensor interfacing, LCD functionality, LED operation, buzzer activation, and stable power distribution across all the components.

Figure 5.2 shows the complete circuit diagram of the proposed system as developed in Cirkit Designer. The DHT11 sensor and MQ Gas Sensor are wired to the Arduino UNO's D4 and A0 pins respectively, the 16×2 I2C LCD is connected through the SDA and SCL lines, and the LED and buzzer are connected to the D8 and D9 output pins along with their associated current-limiting resistor and ground connections.

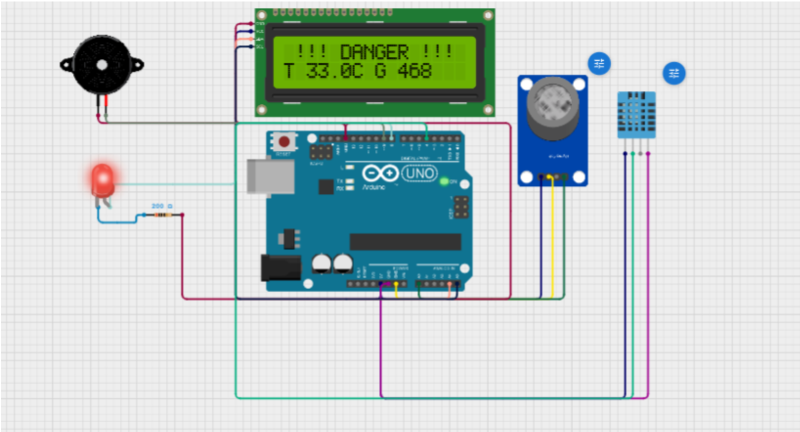

Figure 5.2 Complete Circuit Diagram – Cirkit Designer Simulation

After the simulation results were found satisfactory, the hardware prototype was assembled on a breadboard and tested under real environmental conditions to validate the simulated behaviour against the actual hardware response.

5.3 Experimental Results

The developed system was tested under different environmental conditions by varying the temperature around the DHT11 sensor and exposing the MQ Gas Sensor to combustible gas. Four distinct test cases were carried out, corresponding to the four operating states of the system.

Test Case 1 – Safe Zone

Initially, the system was tested under normal environmental conditions. The observed temperature was 26.4°C and the gas level reading was 235, both of which were below their respective threshold limits. As a result, the LCD displayed SAFE ZONE and both the LED and buzzer remained OFF, as shown in Figure 5.3.

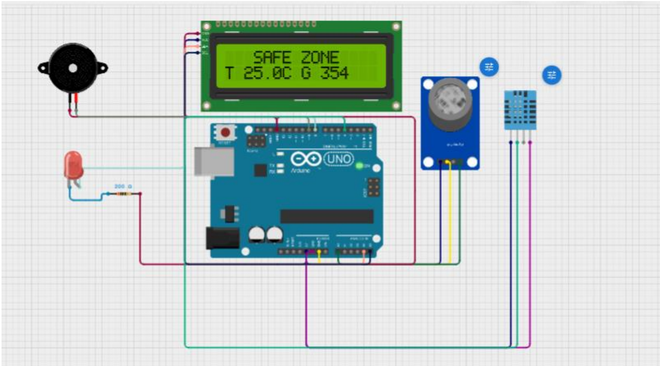

Figure 5.3 LCD Output – Safe Zone

This confirms that the system correctly identifies the normal operating condition and continuously displays sensor values without unnecessarily activating the warning devices.

Test Case 2 – Temperature Warning

The temperature surrounding the DHT11 sensor was gradually increased above the threshold value while the gas concentration was maintained within the normal range. The observed temperature was 31.5°C with a gas level of 228. The LCD displayed TEMP WARNING, while both the LED and buzzer were activated, as shown in Figure 5.4.

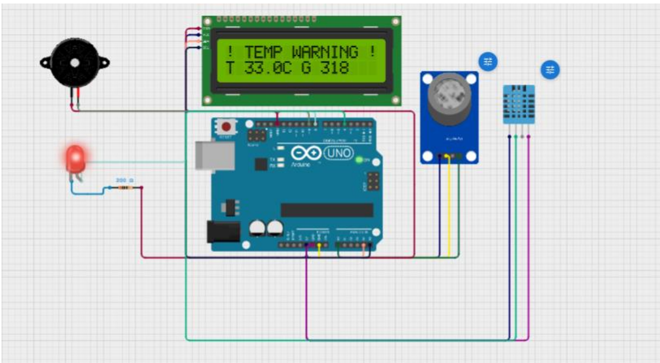

Figure 5.4 LCD Output – Temperature Warning

This test confirms that the system successfully detects abnormal temperature and generates an immediate warning even when the gas concentration remains within the safe limit.

Test Case 3 – Gas Warning

The MQ Gas Sensor was exposed to combustible gas while the surrounding temperature was maintained at a normal level. The observed temperature was 26.2°C with a gas level of 485. The LCD displayed GAS WARNING, and both the LED and buzzer were activated, as shown in Figure 5.5.

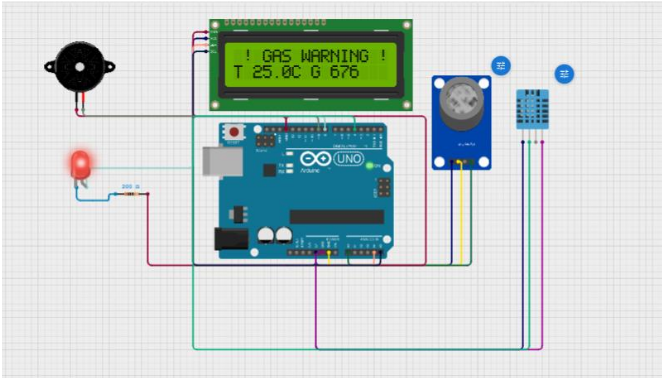

Figure 5.5 LCD Output – Gas Warning

This confirms that the system successfully detects gas leakage and generates an appropriate warning message without being affected by normal temperature conditions.

Test Case 4 – Danger Condition

Finally, both the temperature and gas concentration were increased simultaneously. The observed temperature was 33.0°C with a gas level of 468. The LCD displayed the DANGER message, while both the LED and buzzer remained continuously active, as shown earlier in the circuit diagram of Figure 5.2, which was captured during this danger-condition test.

This test confirms that the system successfully detects simultaneous hazardous conditions and displays the highest priority warning level, alerting the user to the most critical scenario.

5.4 Testing Table

Table 5.1 summarises the results obtained from all four test cases conducted during the experimental evaluation of the system.

Test Case	Temperature	Gas Level	LCD Display	LED	Buzzer	Result
1	26.4°C	235	SAFE ZONE	OFF	OFF	Pass
2	31.5°C	228	TEMP WARNING	ON	ON	Pass
3	26.2°C	485	GAS WARNING	ON	ON	Pass
4	33.0°C	468	DANGER	ON	ON	Pass

Table 5.1 Result Summary of Test Cases

5.5 Discussion

The experimental results demonstrate that the Smart Temperature and Gas Detection System operates reliably under different environmental conditions. The system accurately identifies hazardous situations by continuously monitoring temperature and gas concentration.

Unlike many basic monitoring systems that provide only a general alarm, the developed system distinguishes between temperature-related hazards, gas leakage, and combined danger conditions through separate LCD messages. This feature improves user awareness and enables a quicker response during emergency situations.

Throughout the testing process, the DHT11 sensor provided consistent temperature measurements, and the MQ Gas Sensor responded effectively to changes in gas concentration. The LCD updated the sensor values continuously, and the LED and buzzer reacted immediately whenever hazardous conditions were detected, with no communication delay or hardware malfunction observed. The overall performance of the system indicates that it is well suited for educational demonstrations, laboratory environments, and basic safety monitoring applications.

CHAPTER 6

ADVANTAGES, LIMITATIONS AND FUTURE SCOPE

6.1 Advantages

The developed system provides several advantages over conventional monitoring methods:

- Low-cost implementation using easily available components.
- Simple circuit design that is easy to understand and replicate.
- Easy hardware installation and assembly.
- Real-time monitoring of both temperature and gas concentration.
- Simultaneous detection of multiple hazard conditions.
- Clear visual and audible warning system.
- User-friendly LCD display with distinct status messages.
- Low power consumption suitable for continuous operation.
- Reliable performance verified through repeated testing.
- Easy maintenance and scope for future upgrades.

6.2 Limitations

Although the developed system performs efficiently, a few limitations were identified during the course of this project:

- The DHT11 sensor has limited temperature accuracy compared to more advanced sensors.
- The MQ gas sensor requires proper calibration for precise gas measurement.
- The system is currently intended for indoor use only.
- No battery backup is included in the present design.
- Remote monitoring is not implemented in the current version.
- Automatic emergency response, such as gas valve shut-off, is not available.

These limitations can be addressed through further enhancement of the system in future versions.

6.3 Future Scope

The proposed system can be further enhanced by incorporating advanced technologies. Some of the possible future improvements include:

- Integration with IoT platforms for remote monitoring.
- A mobile application for real-time notifications.
- A GSM module for sending SMS alerts.
- Wi-Fi based cloud data logging.
- A mobile dashboard using platforms such as Blynk or ThingSpeak.
- Automatic exhaust fan control during gas leakage.
- Automatic gas valve shut-off mechanism.
- A voice-based alarm system.
- AI-based hazard prediction using historical sensor data.
- Integration of multiple gas sensors for wider gas coverage.
- Data storage using an SD card module.
- Solar-powered operation for energy-independent monitoring.

These enhancements would significantly improve the functionality, reach, and practical usability of the system in real-world safety applications.

CHAPTER 7

CONCLUSION

The Smart Temperature and Gas Detection System was successfully designed and implemented using an Arduino UNO, a DHT11 Temperature and Humidity Sensor, an MQ Gas Sensor, a 16×2 I2C LCD Display, an LED, and a buzzer.

The developed system continuously monitors environmental conditions and accurately detects abnormal temperature and gas leakage. Whenever the measured values exceed the predefined threshold limits, the system immediately activates visual and audible warning devices while displaying the appropriate warning message on the LCD.

Experimental testing confirmed that the system operates reliably under different environmental conditions. The separate warning messages — SAFE ZONE, TEMP WARNING, GAS WARNING, and DANGER — improve user awareness and enable quick identification of hazardous situations, which is a clear improvement over conventional single-alarm monitoring systems.

The project demonstrates that a simple and low-cost embedded system can effectively provide real-time environmental monitoring and improve safety in homes, laboratories, and small industrial environments. Overall, the developed Smart Temperature and Gas Detection System satisfies the objectives set out at the beginning of the project and provides a reliable foundation for future IoT-based safety monitoring systems.

REFERENCES

[1] Arduino AG, Arduino UNO Rev3 Datasheet. Arduino AG, Somerville, MA, USA, 2021. [Online]. Available: https://www.arduino.cc

[2] Aosong Electronics Co., Ltd., DHT11 Humidity and Temperature Sensor Datasheet. Aosong Electronics, Guangzhou, China, 2014.

[3] Hanwei Electronics Co., Ltd., Technical Data — MQ Gas Sensor Series. Hanwei Electronics, Zhengzhou, China, 2015.

[4] F. de Brabander, "LiquidCrystal_I2C Library," GitHub repository, 2021. [Online]. Available: https://github.com/johnrickman/LiquidCrystal_I2C

[5] Adafruit Industries, "DHT sensor library for ESP8266 and Arduino," GitHub repository, 2023. [Online]. Available: https://github.com/adafruit/DHT-sensor-library

[6] M. Banzi and M. Shiloh, Getting Started with Arduino, 3rd ed. Sebastopol, CA, USA: Maker Media, 2015.

[7] S. Monk, Programming Arduino: Getting Started with Sketches, 3rd ed. New York, NY, USA: McGraw-Hill Education, 2023.

[8] Arduino Software, "Arduino IDE 2 Documentation," Arduino Software Documentation, 2024. [Online]. Available: https://docs.arduino.cc/software/ide-v2

APPENDIX A

ARDUINO SOURCE CODE

```
/*
---------------------------------------------------------
Project Name : Smart Temperature and Gas Detection System
Board        : Arduino UNO
Sensors      : DHT11, MQ Gas Sensor
Display      : 16x2 I2C LCD
Alert System : LED and Buzzer
Author       : Karthikeyan M
Year         : 2026
---------------------------------------------------------
*/

#include <LiquidCrystal_I2C.h>
#include <DHT.h>

// ================= PIN DEFINITIONS =================
#define DHTPIN 4
#define DHTTYPE DHT11
#define MQPIN A0
#define LEDPIN 8
#define BUZZERPIN 9

// ================= THRESHOLDS =================
const int GAS_THRESHOLD = 400;
const float TEMP_THRESHOLD = 28.0;

// ================= OBJECTS =================
LiquidCrystal_I2C lcd(0x27, 16, 2);
DHT dht(DHTPIN, DHTTYPE);

void setup() {
  lcd.init();
  lcd.backlight();
  dht.begin();

  pinMode(MQPIN, INPUT);
  pinMode(LEDPIN, OUTPUT);
  pinMode(BUZZERPIN, OUTPUT);

  Serial.begin(9600);

  lcd.setCursor(0,0);
  lcd.print("  System Ready  ");
  delay(2000);
  lcd.clear();
}

void loop() {
  // ================= SENSOR READ =================
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();
  int gasValue = analogRead(MQPIN);

  // ================= SENSOR ERROR CHECK =================
  if (isnan(humidity) || isnan(temperature)) {
    lcd.setCursor(0,0);
    lcd.print("  Sensor Error  ");
    lcd.setCursor(0,1);
```

```
    lcd.print("  Check DHT11   ");
    Serial.println("DHT11 Error");
    delay(2000);
    return;
  }

  // ================= DANGER LOGIC =================
  bool gasDanger = (gasValue > GAS_THRESHOLD);
  bool tempDanger = (temperature > TEMP_THRESHOLD);
  bool danger = gasDanger || tempDanger;

  // ================= OUTPUT CONTROL =================
  digitalWrite(LEDPIN, danger);
  digitalWrite(BUZZERPIN, danger);

  // ================= DISPLAY LOGIC (FLICKER-FREE) =================
  if (danger) {
    lcd.setCursor(0,0);

    // Condition 1: Both high (Exactly 16 characters)
    if (gasDanger && tempDanger) {
      lcd.print(" !!! DANGER !!! ");
    }
    // Condition 2: Gas high (Exactly 16 characters)
    else if (gasDanger) {
      lcd.print(" ! GAS WARNING !");
    }
    // Condition 3: Temp high (Exactly 16 characters)
    else if (tempDanger) {
      lcd.print("! TEMP WARNING !");
    }

    // Bottom row
    lcd.setCursor(0,1);
    lcd.print("T:");
    lcd.print(temperature, 1);
    lcd.print("C G:");
    lcd.print(gasValue);
    lcd.print("    ");
  }
  else {
    lcd.setCursor(0,0);
    lcd.print("   SAFE ZONE    ");
    lcd.setCursor(0,1);
    lcd.print("T:");
    lcd.print(temperature, 1);
    lcd.print("C G:");
    lcd.print(gasValue);
    lcd.print("    ");
  }

  // ================= SERIAL MONITOR =================
  Serial.print("Temp: ");
  Serial.print(temperature);
  Serial.print(" C | Humidity: ");
  Serial.print(humidity);
  Serial.print(" % | Gas: ");
  Serial.println(gasValue);

  delay(1000);
}
```